

B*-tree floorplan representation

B-star stores packing adjacency: left child sits right-of the parent; right child sits above on a contour

Pseudocode

- B-star packing walks the tree with a contour
- Left child sits to the right of the parent
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Golden tree roots A at the origin
- B's x equals A's right edge; D's y is at least A's height
- The packing must stay legal in the outline

Algorithm sketch — try these

```
INPUT: binary tree (left=right-of, right=above)
OUTPUT: packed (x,y) via contour
root at (0,0)
left child:  $x \leftarrow \text{parent}.x + \text{parent}.w$ 
right child:  $y \leftarrow \text{above parent (contour)}$ 
update horizontal contour after each place
GOLDEN:  $A@0,0$ ;  $B.x=A.x+A.w$ ;  $D.y \geq A.h$ ; legal
```

Root A at the origin

B*-TREE

Root A at the origin

Step 1 / 5 · root · module02-03-bstar-tree

10×8



Explanation

B-star packing places the root at the lower left. A lands at zero comma zero with size three by two. Left children go right-of; right children go above via the contour.

- Root = lower-left
- Left child = right-of parent
- Right child = above on contour

A at (0,0)
tree root: A

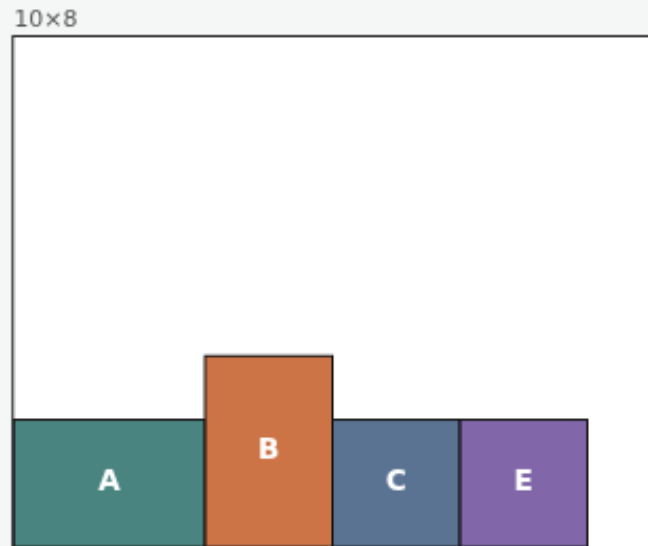
Root A at the origin

Left chain $B \rightarrow C \rightarrow E$

B*-TREE

Left chain $B \rightarrow C \rightarrow E$

Step 2 / 5 · left-chain · module02-03-bstar-tree



Explanation

The left spine walks rightward: B at x equals three, then C, then E. Contour heights track the skyline so later modules sit tightly.

- $B.x = A.x + A.w = 3$
- C and E continue right
- Contour updates each place

B at (3,0)

C at (5,0)

E at (7,0)

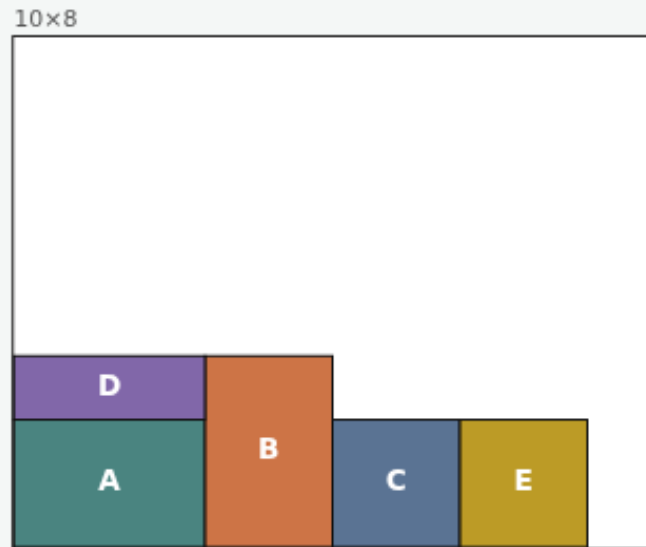
Left chain $B \rightarrow C \rightarrow E$

Right child D above A

B*-TREE

Right child D above A

Step 3 / 5 · right-d · module02-03-bstar-tree



Explanation

D is the right child of A, so it packs above A on the contour. Its y is at least two—A's height—keeping the tree geometry honest.

- $D.y \geq A.h$
- Same x as A branch
- Still non-overlapping

D at (0,2)
A.h=2

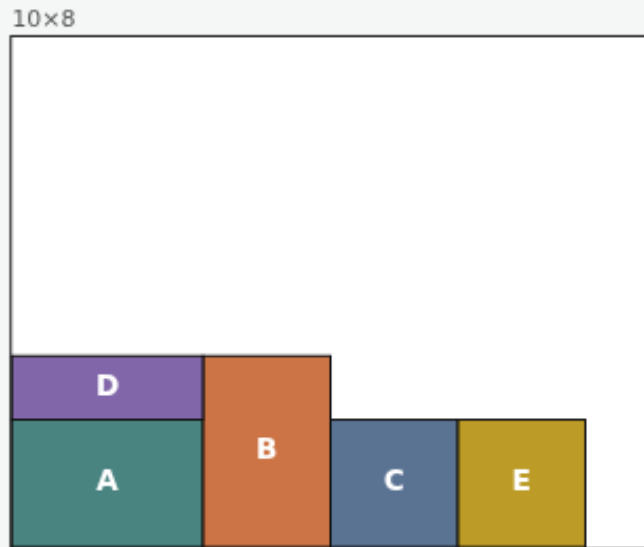
Right child D above A

Full B* pack is legal

B*-TREE

Full B* pack is legal

Step 4 / 5 · legal · module02-03-bstar-tree



Explanation

All five modules are placed. The packing is legal inside ten by eight. Perturbing the tree—swap, rotate, move—will feed simulated annealing later.

- Five modules placed
- legal: true
- Tree is a packing code, not netlist

```
legal: true  
modules: A-E
```

Full B* pack is legal

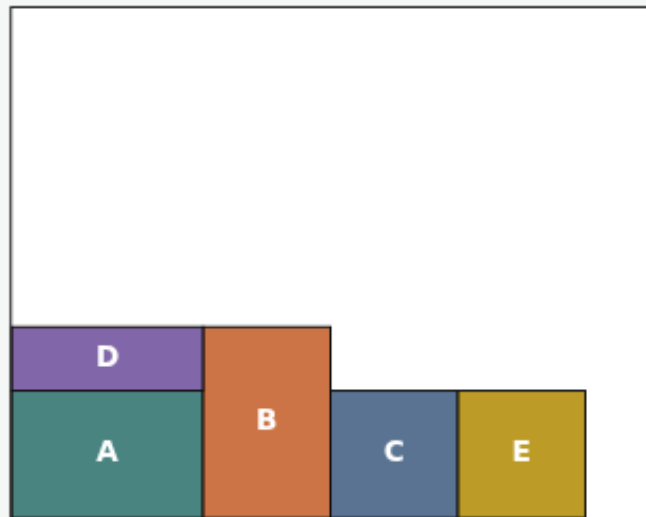
B* is compact and mutable

B*-TREE

B* is compact and mutable

Step 5 / 5 · takeaway · module02-03-bstar-tree

10×8



Explanation

B-star stores adjacency for packing, not connectivity. Get left and right semantics right, keep the contour correct, and you have a fast neighbor generator for search.

- Left/right geometry is the bug magnet
- Contour must update on place
- Next: sequence-pair alternative

starter golden: legal B* pack

B* is compact and mutable

Browser lab track

- Open bstar-tree and Pack B*-tree
- Confirm A at zero comma zero, B at x equals three, D above A

Implement track

- Build the golden tree and contour-pack it
- Assert A at (0,0), B.x equals A.x plus A.w, D.y at least A.h, and legality true

Pitfalls

- Reversing left/right geometry; stale contour segments; treating the tree as a netlist

Your turn

- Produce the legal B-star packing
- Next: sequence-pair permutations as another encoding